

SAATHI — Team 3

Complete RAGA/LLM, RAG, Grounding, Citation and Multilingual AI Work Package

Operating rules

- Keep the existing ingestion package, versioning package, current AI/backend, and the ~1.25 GB RAGA model. Do not delete or retrain existing assets before benchmarking them.
- Use the existing 600+ MB government-source dataset as the controlled starting corpus. Every source must retain provenance; do not silently add unverified web data.
- Work in separate Git branches and reproducible Colab/Kaggle notebooks. Every experiment must record dataset/model/index version and metrics.
- Never replace a working baseline without a measured comparison.

Mission

- Make the existing RAGA model useful as a controlled evidence-to-answer model, and improve answer quality without blindly fine-tuning. First identify exactly what the 1.25 GB model is and how it is intended to run.

Step 1 — Model identification

- Record exact model name, architecture, parameter count, tokenizer, context length, file format, quantization, intended task and hardware requirements.
- Determine whether the 1.25 GB asset is an LLM, embedding model, reranker, classifier or another component. Do not assume from the file size or the word RAGA.
- Create a model card and a reproducible loading notebook.

Step 2 — Baseline model evaluation

- Run the existing model on a fixed set of BIS questions with supplied evidence.
- Measure answer correctness, citation correctness, unsupported-claim rate, refusal behavior and latency.
- Keep the unmodified model as the baseline.

Step 3 — Evidence-conditioned RAG

- Accept Team 2's top evidence chunks and metadata.
- Construct a strict prompt containing question, evidence, language and answer rules.
- Test context size, evidence ordering and number of chunks; retain settings based on evaluation rather than intuition.

Step 4 — Grounding and cite-or-decline

- Require the model to answer only from supplied evidence.
- Detect unsupported claims and either regenerate with reduced claims or decline.
- Attach citations from retrieved metadata; never allow the model to invent source URLs, pages or clauses.

Step 5 — Answer quality improvement

- Build a failure taxonomy: missing evidence, wrong evidence, over-answering, citation mismatch, version confusion, false-premise acceptance and language failure.
- Improve retrieval/context/prompting before considering fine-tuning.
- Use structured answer templates for regulatory claims: claim, evidence, explanation, action, uncertainty.

Step 6 — English/Hindi

- Test English→English, Hindi→Hindi, Hindi→English and English→Hindi.
- Test transliterated Hindi queries.
- Keep standard numbers, clause numbers and citations unchanged across languages.

Step 7 — Targeted model improvement

- Only if evaluation shows a consistent model-specific failure after retrieval and grounding are strong, test LoRA/PEFT or a task-specific classifier/reranker.
- Use train/validation/test separation and compare against the frozen baseline.
- Do not fine-tune the foundation model merely to make the project look more AI-heavy.

Step 8 — Safety and uncertainty behavior

- Test unknown/out-of-domain questions, false premises, similar standards and outdated standards.
- Required behavior for unsupported regulatory claims: explicit uncertainty or decline.
- Never convert missing evidence into a confident compliance requirement.

Deliverables

- model card
- model loading notebook
- RAG pipeline
- prompt/version registry
- grounding validator
- citation mapper
- English/Hindi evaluation report
- optional fine-tuning experiment only if justified

Acceptance test

- Model answers evidence-supported questions correctly.
- Unsupported questions produce a controlled decline.
- Citations point to retrieved evidence.
- Changing the model prompt cannot manufacture source metadata.